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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - APRIL 2021

CH306S - ANALYTICAL CHEMISTRY - II

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. The Property measured in thermogravimetric analysis is
   a) Change in weight b) Rate of Change in weight
   c) Heat Evolved or absorbed d) Change of temperature
2. The Process of converting precipitate into colloidal form by shaking it with dispersion medium in the presence of small amount of electrolyte
   a) Co precipitation b) Post –precipitation
   c) Peptisation d) None of these
3. ---------- is a method of distillation performed under reduced pressure
   a) Steam distillation b) Vaccum distillation c) Crystallization d) Sublimation
4. ---------- method is used in the separation of proteins
   a) Franctional crystallization b) Franctional distillation
   c) Vaccum distillation d) Electrophoresis
5. Ilkovic equation is
   a) $id= 708\ nCD^{1/2} m^{2/3} t^{1/6}$ b) $id= 708\ nCD^{1/6} m^{2/3} t^{1/2}$
   c) $id= 706\ nCD^{1/2} m^{2/3} t^{½}$ d) $id= 708\ nCD^{2/3} m^{2/3} t^{1/6}$
6. In amperometric titrations the measured current is
   a) Directly proportional to concentration of analyte
   b) Direnctly proportional to applied potential
   c) Indirectly proportional to concentration
   d) Inversely proportional to applied potential
7. An absorption band around 170-190 nm is observed in unconjugated alkenes due to the transition.
   a) n- σ* b) n-σ* c) n-Л d) n-Л*
8. Bragg's equations is
   a) $n\lambda=2d \sin_{\Pi} o$ b) $n\lambda=2d \cos 0$ c) $n\lambda=2d \tan o$ d) $n\lambda=2d \sin\theta$
9. Sea water, river water and tap water are
   a) Hard water b) Soft water c) Both a&b d) Distilled water
10. The process of destroying the harmful bacteria is
    a) Sterilization b) Coagulation c) Sedimentation d) Filtration

II. Fill up the blanks

11. _____is the simultaneous precipitation of more then one compound from a solution.
12. _____is the separation of liquid mixture into fractional differing in boiling point.
13. DME is used in _____
14. _____ transition is involved in uv-visible spectroscopy.
15. BOD means _____

II. Match the following

16. DTA	- Absorption
17. Steam distillation	- Water treatment
18. Polarography	- Thermal analysis
19. Beer Lambert law	- Purification of liquid
20. lime soda process	- Current voltage curve

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. What is meant by specific precipitants?
22. Write any two factors affecting TGA and DTA curves?
23. Define Electrophoresis.
24. What is meant by sublimation?
25. Write any two advantages of DME?
26. Write any two uses of amperometry.
27. What is diffusion current?
28. State Beer's law.
29. Write the structure of NaCl.
30. What is permanent hardness?
31. What is sedimentation?
32. What is COD?

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a) Discuss about co-precipitation and post precipitation. **(4)**
    b) What is sequestering agent? Give example? **(3)**
34. a) Write a Block diagram of TGA. **(4)**
    b) Write a note on DTA curve. **(3)**
35. a) Write a note on vacuum distillation. **(4)**
    b) Write a note on crystallization. **(3)**
36. a) Explain the significance of illkovic equation? **(4)**
    b) Write a note on polarography. **(3)**
37. a) Explain the instrumentation of photo calorimeter. **(4)**
    b) Write a note on powder technique. **(3)**
38. a) Write a note on reverse osmosis. **(4)**
    b) How to calculate temporary and permanent hardness. **(3)**
39. a) Write a note on lime soda process. **(4)**
    b) What is coagulation? **(3)**

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